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Unlock Your Heart *Health*

The Cholesterol, Blood Pressure, and Fatty Liver Connection Nobody Draws For You

A guide for adults who just got lab numbers or a blood pressure reading back that are trending the wrong way, who have not had a heart attack or stroke, and who want to understand what is actually driving those numbers instead of just being told to watch their diet.

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This edition is pending Dr. Seay's clinical sign-off. Every citation has been verified against its live PubMed record, including a check for retractions. The companion citation ledger lists each source and the claims that were investigated but left uncited.

BEFORE YOU START

A note before you start. This guide is educational information about the biology of cardiometabolic risk (how high cholesterol, high blood pressure, and fatty liver relate to each other) and the general categories of lifestyle and functional-medicine approaches used to address the underlying drivers. It is not individualized medical advice, and it does not replace a relationship with your physician or a clinical evaluation with me and my team. It does not diagnose anything. If you already have diagnosed heart disease, have had a heart attack, stroke, or stent or bypass procedure, or have diagnosed kidney disease, this guide is background reading only. Your cardiologist or nephrologist directs your care, and a direct conversation with our team is a better next step for you than working through this guide on your own. If you are currently taking a statin, a blood pressure medication, or a blood thinner, keep taking it exactly as prescribed. Nothing in this guide is a reason to stop, skip, or reduce any prescription medication. Talk to your prescribing physician before starting any new supplement, especially if you take any of the medications above, are pregnant or nursing, or have kidney disease.

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Before Anything Else: When To Stop Reading And Call 911

This guide is written for someone with rising numbers, not someone in the middle of an emergency. Please read this section once, know it, and come back to the rest of the guide when it applies to you.

Call 911 or get to an emergency room immediately if you experience:

- Chest pain, pressure, squeezing, or tightness, especially if it lasts more than a few minutes or comes and goes
- Pain, pressure, or discomfort spreading to the jaw, neck, back, shoulder, or one or both arms
- Sudden shortness of breath, with or without chest discomfort
- Sudden weakness or numbness on one side of the face, arm, or leg, facial drooping, or sudden trouble speaking or understanding speech (the classic warning signs of stroke, sometimes taught as F.A.S.T.: Face drooping, Arm weakness, Speech difficulty, Time to call 911)
- A blood pressure reading of 180 systolic or higher, or 120 diastolic or higher (a hypertensive crisis range), particularly if paired with chest pain, shortness of breath, back pain, numbness, weakness, vision change, or difficulty speaking
- Sudden severe headache unlike any you have had before, sudden vision loss, or sudden loss of balance or coordination
- Nausea, cold sweat, or lightheadedness together with chest or upper body discomfort

None of these symptoms are something to sit with, look up, or wait out until your next appointment. They are something to act on immediately. This section exists because this guide will be read by people at genuine cardiovascular risk, and knowing these signs matters more than anything else in this document.

Everything below this point is for a different situation: understanding and addressing risk that is building quietly, before any of the above happens.

INTRODUCTION

The Numbers Nobody Connects For You

Maybe your cholesterol panel came back with a red flag next to LDL. Maybe your blood pressure has been creeping up at your last few checkups, and someone handed you a pamphlet about cutting salt. Maybe an ultrasound or CT scan done for something else mentioned "fatty liver" almost in passing, and nobody explained what that meant or why it was there.

Here is what usually does not happen in a fifteen minute visit: nobody sits down and shows you how those three things, high cholesterol, high blood pressure, and a fatty liver, are frequently the same underlying problem showing up in three different places. They get treated as three separate conversations, sometimes with three separate specialists, when in a large number of people they trace back to one shared root: how your body is handling insulin and blood sugar at the cellular level.

That is not a reason to distrust your cholesterol medication or your blood pressure prescription. Those medications treat real, measurable risk, and later in this guide we spend real time on exactly why that is true and why continuing them matters. It is a reason to understand what else is going on underneath the numbers, so that the terrain producing those numbers gets addressed too, alongside the medications, not instead of them.

This is the cluster of conditions that, taken together, is responsible for more deaths than almost anything else in modern medicine. That is precisely why it deserves a clear-eyed, non-alarmist explanation rather than either dismissal or panic. Below are nine keys. Each one explains a piece of the cardiometabolic picture, the mechanism behind it in plain language, and what you can do about it at an educational level. This guide does not diagnose you and it is not a treatment plan. It is the explanation that usually gets skipped.

KEY 1

Insulin Resistance Is the Shared Root of All Three

Three problems, one root. High cholesterol, high blood pressure, and a fatty liver are usually discussed as three unrelated problems that happen to show up in the same person as they age. In a large share of cases, they are not unrelated at all. They are three downstream expressions of the same upstream problem: cells that have stopped responding normally to insulin.

How one signal drives three outcomes. Insulin's job is to help your cells take up glucose from your blood. When cells become resistant to insulin's signal, most commonly driven by excess adiposity and a chronically high intake of refined carbohydrates, your pancreas compensates by producing more insulin to force the message through. Researchers describe insulin resistance as a root causative factor not only for metabolic syndrome itself but for the associated conditions that ride along with it: fatty liver disease, obesity-related type 2 diabetes, and atherosclerotic cardiovascular disease.[1] There is an active scientific debate about which comes first: whether insulin resistance itself drives the downstream damage, or whether the earlier problem is actually too much insulin (hyperinsulinemia), with insulin resistance developing afterward as a protective adaptation by tissues like the heart muscle trying to defend themselves against insulin overexposure.[1] Both camps agree on the practical point that matters here: disrupted insulin signaling is the mechanistic thread connecting dyslipidemia, elevated blood pressure, and fatty liver, not three separate coincidences.

Mechanistically, chronically elevated insulin pushes the liver to manufacture more triglyceride-rich particles (raising triglycerides and driving the small, dense LDL pattern discussed in Key 2), promotes sodium retention and sympathetic nervous system activation in the kidney and blood vessels (contributing to the blood pressure story in Key 6), and drives fat storage directly inside liver cells (the fatty liver story in Key 5). One disrupted signaling pathway, three visible symptoms.

How strong is that claim, really. This is the part worth pausing on, because "insulin is the root cause" is said constantly in wellness circles without anyone showing you the evidence, and it deserves better than a slogan.

The strongest available evidence here is not a diet study. It is a technique called Mendelian randomization, which uses the gene variants you were born with as a natural experiment. Because those variants are assigned at conception and cannot be influenced by your diet, your income, or your habits, they sidestep most of the confounding that makes ordinary nutrition research so difficult to interpret. It is about as close to a randomized trial as you can get for something nobody could ethically randomize.

Applied to insulin resistance, the answer is striking. Using 53 gene variants tied to the insulin-resistant phenotype, drawn from studies of up to 188,577 people for the exposure and 446,696 for the outcome, a one standard deviation increase in genetically predicted insulin resistance was associated with a **79 percent higher risk of coronary artery disease** (odds ratio 1.79), a 78 percent higher risk of myocardial infarction, and a 21 percent higher risk of ischemic stroke, with no evidence of the statistical pleiotropy that would undermine the finding. The authors concluded that insulin resistance is causally associated with coronary artery disease and myocardial infarction.[2] A separate 2025 analysis using different data and a different method arrived at almost exactly the same effect size for fasting insulin, an odds ratio of 1.79 for coronary artery disease, and went further by identifying the specific circulating proteins carrying that effect.[3]

So the claim that insulin sits upstream of cardiovascular disease is not a wellness talking point. It is supported by the same class of causal evidence used to establish the cholesterol relationship itself.

And here is where honesty requires stopping. A sex-specific analysis of 392,010 people in the UK Biobank found the association between genetically predicted insulin and both myocardial infarction and angina held **in men but not in women**.[4] That is one study and it does not overturn the others, but a guide that quotes the good numbers and hides that one is not worth your money. If you are a woman reading this, the insulin work is still worth doing for every other reason in this book, and the cardiovascular claim specifically is less settled for you than the headline suggests.

Insulin versus cholesterol is a false fight. This is the most useful thing in this chapter. The two positions get argued as though only one can be true, and they are actually the same story told at two different points in the same chain.

When researchers compared the lipid measurements head to head using that same genetic method, **apolipoprotein B was the one that survived**. Analyzed individually, LDL cholesterol, triglycerides and ApoB all looked harmful. Analyzed together, only ApoB retained a robust effect, the estimate for LDL cholesterol collapsed and even reversed direction, and the apparent protective effect of HDL attenuated essentially to nothing once ApoB was accounted for. The conclusion was that ApoB is the predominant trait explaining the relationship between blood lipids and coronary heart disease.[5] ApoB, as Key 2 explains, is a count of atherogenic particles rather than a measure of the cholesterol carried inside them.

Now put the two findings side by side. Insulin resistance drives the liver to manufacture more of exactly those particles. And in the proteomic analysis above, one of the mediators identified between insulin and coronary artery disease was the LDL receptor pathway itself.[3] **Insulin resistance is upstream. Particle number is the mechanism. Arterial damage is the outcome.** Anyone telling you it is one or the other is defending a position rather than describing the biology.

Which also makes the honest boundary of this chapter clear: insulin resistance is not the only road in. Familial hypercholesterolemia and elevated Lp(a) are genetically determined, they raise particle-driven risk in people who are lean and insulin sensitive, and no amount of metabolic work removes them. That is precisely why Key 2 asks you to measure them.

So why has nobody measured your fasting insulin. If this is genuinely upstream, the obvious question is why it is not on the standard panel. The reasons are structural rather than sinister, and knowing them helps you ask better questions.

First, **the test itself is not standardized between laboratories.** Insulin immunoassays show significant relative differences from one platform to another, and work on harmonizing them found that recalibration improved agreement at middle and high insulin levels **but not at low ones**, which is exactly the range where early detection would matter most.[6] A test without a universal number is a test that guidelines struggle to set a threshold for.

Second, **the diagnostic categories are built on glucose, not insulin.** Prediabetes and diabetes are defined by blood sugar and HbA1c. As described above, insulin climbs for years to hold glucose normal, so a system that screens on glucose is calibrated to find the problem only after the compensation has begun to fail.

Third, **fasting insulin is not in the cardiovascular risk calculators.** The equations your physician uses to make prevention decisions take cholesterol, blood pressure, smoking, age, and diabetes status. A marker that is not in the equation cannot change the estimate, and a marker that cannot change the estimate does not change the recommendation, however real it is.

Fourth, **the three downstream problems each have an owner and the upstream one does not.** Lipids, blood pressure, and the liver belong to different specialties, each with its own guideline and its own drug. Insulin resistance sits underneath all three and is nobody's assigned territory.

And fifth, worth saying plainly given how much of this book rests on published trials: **no company profits from proving that your insulin sensitivity improved.** Drug trials get funded because someone can sell the result. That asymmetry does not make the drug evidence wrong, and this book cites a great deal of it. It does mean the evidence base is far thicker on one side of this comparison than the other, and you should read that imbalance as a fact about how research gets funded rather than as a fact about your body.

None of this is an argument against treating cholesterol or blood pressure. It is an argument for asking one more question at your next appointment.

Where this points you next. This is why addressing insulin sensitivity, through nutrition pattern, movement, sleep, and in some cases targeted nutrient support, tends to move all three numbers at once rather than requiring three separate unconnected interventions. The rest of this guide walks through each expression of that shared root in more detail, and Key 9 covers the nutrition and movement patterns with real evidence behind improving insulin sensitivity itself.

KEY 2

What Your Standard Lipid Panel Does Not Tell You

What the same LDL number can hide. A standard cholesterol panel reports total cholesterol, LDL, HDL, and triglycerides. That panel has real value, and nothing in this guide is a reason to think otherwise. But two people can have the identical LDL number on paper and carry meaningfully different cardiovascular risk, because LDL cholesterol is a measurement of the cholesterol cargo inside a set of particles, not a direct count of the particles themselves.

What ApoB actually counts. Every atherogenic particle (the ones capable of lodging in your artery wall) carries exactly one molecule of a protein called apolipoprotein B (ApoB). That makes ApoB a direct count of the number of particles in circulation, rather than an estimate of the cholesterol they happen to be carrying that day. Because the amount of cholesterol packed into each particle varies from person to person, someone can have many small, cholesterol-poor particles (a high ApoB, higher risk) while their calculated LDL-C looks unremarkable, or the reverse.[7] This is exactly why professional cardiology and lipid societies increasingly reference ApoB alongside, not instead of, standard LDL-C: it resolves cases where the standard panel and the true particle burden disagree.

Lipoprotein(a), often written Lp(a), is a distinct, LDL-like particle that carries its own additional protein tag, making it more prone to promoting both plaque buildup and clotting. It is measured on request, not as part of a standard panel. Roughly seventy to ninety percent of the variation in Lp(a) levels between people is genetically determined, which means it is one of the few cardiovascular risk markers that is essentially set from an early age and does not move much with diet or lifestyle.[8] It also does not come down with statin therapy, since statins work on a different pathway. That does not mean it is unmanageable information, it means it changes how aggressively the other, modifiable risk factors (ApoB, LDL-C, blood pressure, inflammation) are worth managing in someone who has it.[8] Lp(a) is worth checking once, since it does not meaningfully change over your lifetime once you are past early childhood.

A third figure worth knowing is your triglyceride-to-HDL ratio, calculated from numbers already on a standard panel. It functions as an inexpensive stand-in for insulin resistance itself, the shared root from Key 1. Research reviewing decades of data across tens of thousands of people found average cutoffs of roughly 2.5 for women and 2.8 for men, above which insulin resistance becomes substantially more likely, though the exact cutoff varies somewhat by ethnicity and sex.[9]

And a fourth number, homocysteine, deserves the most honest handling of anything in this chapter. Homocysteine is an amino acid your body produces and clears as part of the methylation cycle, a set of reactions that runs on vitamin B12, folate, and B6. When one of those vitamins runs short, when kidney or thyroid function is off, or when a common gene variant (MTHFR) slows the recycling enzyme, homocysteine backs up in the blood. That is what makes it genuinely worth measuring: it is an inexpensive window into B-vitamin status, kidney function, and methylation capacity, systems worth knowing about in their own right.

What it is not, on the best current evidence, is a cardiac number to chase for its own sake, and the story of how we know that is the mirror image of the statin story in Key 3. In observational data pooling 30 studies, people with lower homocysteine did have modestly lower risk, about 11 percent less ischemic heart disease and 19 percent less stroke per 3 umol/L difference, and even the researchers reporting that association called homocysteine “at most a modest independent predictor” of those outcomes.[38] Then two harder tests arrived. A Mendelian randomization analysis, the same genetic technique this book leans on for insulin and ApoB, deliberately gathered 19 unpublished datasets totaling 48,175 coronary cases and found that lifelong moderately elevated homocysteine, produced by the MTHFR variant, had little or no effect on coronary heart disease: an odds ratio of 1.02, statistically indistinguishable from no effect. The previously published literature showed a significant 1.15, and the authors attributed the gap to publication bias, meaning small positive studies got published while null results stayed in file drawers.[39] And when randomized trials actually lowered homocysteine with folic acid in more than 82,000 people, stroke fell by about 10 percent and overall cardiovascular events by about 4 percent, with the benefit concentrated in people who started out folate-deficient, while coronary heart disease did not budge.[40]

Notice what happened there, because it is the same lesson as Key 3 running in the opposite direction. The published homocysteine literature was biased toward the position wellness culture favors, exactly as industry funding tilted the statin literature toward the position drug manufacturers favor. This book applies one standard to both: the strongest evidence wins, whoever it disappoints.

So here is the working conclusion, and it is the same one we use clinically at Root Health: treat an elevated homocysteine as a signal to investigate B12, folate, B6, thyroid, and kidney status, not as a target proven to change cardiac outcomes on its own. A result above roughly 10 umol/L is worth that workup conversation with your physician. Functional medicine practices, ours included, often read against a tighter target near 7, and standard laboratory ranges run looser still, so ask which frame your result is being read against. Two boundaries on all of this. The genetic analysis above speaks only to moderate, lifelong elevation;[39] homocystinuria, the rare inherited disorder where levels run several times normal, is a different and genuinely dangerous condition that belongs under medical management. And if the workup does find a B12 or folate deficiency, correcting it matters for your nerves, your blood, and your brain regardless of what it does or does not do for your arteries.

What these numbers change. None of this means your standard lipid panel was wrong, or that LDL cholesterol does not matter. It means there are more precise ways to measure the same underlying risk, and functional medicine uses ApoB, Lp(a), the triglyceride-to-HDL ratio, and homocysteine to sharpen the picture your standard panel already started, not to replace it or argue it was mistaken. If you already know your LDL is elevated, these additional numbers help clarify how urgently and how the modifiable pieces need to be addressed, working alongside whatever your prescriber has already recommended.

KEY 3

What the Statin Evidence Actually Shows, and What It Does Not

Two loud extremes, both wrong. Statin messaging tends to arrive from two opposite extremes. One says the entire cholesterol story is a myth and you should quietly stop your prescription in favor of supplements. The other treats "you should be on a statin" as settled for nearly everyone with an elevated number, full stop. Neither extreme reflects the actual evidence. What the data shows, how strong it is, and what it means for you depends heavily on one specific fact: whether you have already had a cardiac event, or you are looking at a number trending the wrong way for the first time. Collapsing those two situations into one blanket answer is exactly what makes this topic feel more settled, or more suspect, than it actually is.

Where genetics and drug trials agree. Start with what is genuinely not in dispute. LDL cholesterol's role in causing atherosclerotic cardiovascular disease does not rest on drug trials alone. Mendelian randomization studies, which use naturally occurring genetic variants that lower LDL from birth as a kind of lifelong natural experiment immune to industry funding or early trial stopping, consistently find the same result as the drug trials: lowering LDL, by whatever mechanism, lowers cardiovascular risk in proportion to how much it is lowered and for how long it stays lowered. A 2017 consensus statement from the European Atherosclerosis Society, reviewing genetic, epidemiologic, and clinical trial evidence together, concluded plainly that this combined evidence "unequivocally establishes that LDL causes ASCVD."[28] That is why "cholesterol has nothing to do with heart disease" is not a defensible reading of the literature. It is a separate question from how large a benefit a specific person, at a specific risk level, actually gets from taking a statin, which is where the real nuance lives.

The broadest randomized evidence for statin therapy comes from a meta-analysis pooling individual data from over 174,000 participants across 27 trials, which found major vascular events fell by about 22 percent in men and 16 percent in women per 1.0 mmol/L LDL reduction, with reduced all-cause mortality in both sexes and no signal of increased cancer or non-cardiovascular death in either sex.[10] That trial population is a mix that skews toward higher average risk than a reader who is simply noticing a rising number for the first time, including a substantial share of people who already have documented vascular disease, which is exactly why the absolute size of that 22 and 16 percent benefit needs to be read through the two categories below rather than as one number that applies equally to everyone: first the concrete absolute numbers for people who already have cardiovascular disease, then for people who do not.

If you have already had a heart attack, a stroke, or a stent or bypass procedure, the evidence is at its strongest and least contested. This is called secondary prevention: treating someone who already has documented disease, rather than trying to prevent a first event. In the trial that first established this, 4,444 patients with angina or a prior heart attack were randomized to simvastatin or placebo and followed for a median of 5.4 years. In the placebo group, 12 percent died; in the simvastatin group, 8 percent died, a 4 percentage point absolute difference (relative risk 0.70, a 30 percent reduction). Major coronary events occurred in 28 percent of the placebo group versus 19 percent of the simvastatin group, a 9 percentage point absolute difference (relative risk 0.66, a 34 percent reduction).[29] One more fact belongs in the same breath, because this guide checks funding in every direction: that trial was funded by Merck, the manufacturer of simvastatin, as nearly every landmark statin trial was manufacturer-funded. The reason the finding stands anyway is that the same direction of benefit shows up in the pooled analyses run by publicly funded academic groups[10] and in the genetic evidence described above, which no company sponsored.[28] A difference of several percentage points, measured over a few years, in people who already have the disease, is what strong evidence looks like in this field. If this is your situation, staying on your statin is not a close call.

If you have not had a cardiac event and are simply watching a number trend upward, the picture changes, not because the drug works differently, but because your starting risk is much lower, so the same relative benefit produces a much smaller absolute one. A 2012 meta-analysis from the Cholesterol Treatment Trialists' Collaboration, examining its pooled trial dataset specifically in people at low baseline vascular risk, found the proportional risk reduction per 1.0 mmol/L LDL reduction was at least as large in the lowest-risk participants as in higher-risk ones (a relative risk of 0.62 in the very lowest risk category, meaning roughly a 38 percent reduction), genuinely good news on the relative side. Translated into absolute terms, though, for people with a 5-year risk of a major vascular event under 10 percent, that reduction worked out to about 11 fewer events per 1,000 people treated over 5 years.[30] Eleven people out of a thousand, over five years, is a real benefit. It is also a small one, smaller than a 30 or 38 percent relative-risk figure tends to convey on its own. That is the actual, honest size of the number a low-risk reader is weighing, and well-informed people reasonably land in different places once they see it stated this way, which is exactly why this is an individual conversation with your prescriber rather than a foregone conclusion in either direction.

This same relative-versus-absolute gap is why one widely cited statin trial deserves closer scrutiny than it usually gets. That trial enrolled people with already-low LDL cholesterol but elevated hs-CRP (Key 4), and it is commonly summarized as a 47 percent reduction in first heart attack, stroke, or cardiovascular death, a figure repeated often enough that it functions as shorthand for "statins work."[11] The actual event rates behind that percentage were 0.77 per 100 person-years in the statin group versus 1.36 per 100 person-years in the placebo group, an absolute difference of about six-tenths of one percentage point per year.[31] That is a real effect, not a null one. But three things about this specific trial matter before it gets treated as a pillar of the case for statins. It was stopped early on a prespecified rule after a median of only 1.9 years of follow-up, and trials stopped early on their most favorable-looking interim result tend to overstate the true effect size. Independent reviewers have specifically pointed to the trial's strong commercial sponsorship as a plausible source of bias in how it was conducted and interpreted.[32] And the trial's lead investigator has disclosed being a co-inventor on patents related to the inflammatory biomarker testing used to select who could enroll in the first place.[33] None of that makes the finding false. It means this particular trial is the weakest possible anchor for the broader statin case, and a 47 percent headline without the underlying rate or these caveats is precisely the kind of framing that fuels distrust, and on this specific trial, that distrust has a real basis.

Two more things deserve honest treatment here, because patients ask about them before almost anything else, and a guide that leaves them out is not giving you the full picture. First, statins measurably raise the risk of new-onset type 2 diabetes. This is not a fringe claim. It is in the FDA-required product label, and it comes from the statin trialists' own pooled data. A meta-analysis of 13 trials and more than 91,000 participants found statin therapy carried a 9 percent increased relative risk of incident diabetes (odds ratio 1.09), which worked out to treating 255 people with a statin for 4 years to produce one additional case of diabetes.[34] A larger 2024 analysis of individual participant data from the same collaboration, covering 19 placebo-controlled trials and nearly 124,000 people, found low or moderate intensity statin therapy carried a 10 percent proportional increase in new-onset diabetes (1.3 percent of people affected per year versus 1.2 percent on placebo, a difference of one-tenth of one percentage point per year), while high-intensity statin therapy carried a substantially larger 36 percent proportional increase (4.8 percent per year versus 3.5 percent, a difference of 1.3 percentage points per year).[35] This is a genuine, dose-related effect, largest at the highest statin doses. It cuts the opposite direction from how it sometimes gets used in arguments: a drug that nudges blood sugar the wrong way is a reason the terrain-level work in the rest of this guide, insulin sensitivity, visceral fat, sleep, and nutrition, matters more, not less, alongside the medication rather than instead of it.

Second, muscle aches and pains are the single most common reason people say they want off a statin, and the honest answer is that trial data and real-world experience genuinely disagree with each other in a way that is not fully resolved. A 2019 American Heart Association scientific statement on statin safety found that across randomized, placebo-controlled trials, the difference in muscle symptoms between statin-treated and placebo-treated participants was under 1 percent, and under 0.1 percent for symptoms severe enough to stop the drug, while noting that roughly 10 percent of patients discontinue a statin in ordinary clinical practice, overwhelmingly citing muscle complaints as the reason.[36] A 2021 trial published in the BMJ tested this directly: 151 patients who had previously stopped a statin because of muscle symptoms were rotated, blinded, through repeated statin and placebo periods, and found no measurable difference in muscle symptom scores between the statin periods and the placebo periods.[37] Both findings are true at once: trial-measured rates are low, real-world discontinuation rates are meaningfully higher, and researchers do not fully agree on how much of that gap reflects a true drug effect trials underestimate versus expectation and attention effects that grow once someone knows what they are taking. If you have had muscle symptoms on a statin, that experience is real and worth a specific conversation with your prescriber, including whether a different statin, a different dose, or a supervised trial off and back on makes sense. It is a decision for that conversation, not one to make alone.

Put plainly, here is what is settled and what is not. Settled, and not seriously contested among researchers: LDL's causal role in cardiovascular disease, the strength of secondary-prevention benefit for people who have already had a cardiac event, and the diabetes-risk effect described above. Genuinely contested, or still being actively debated: the true size of absolute benefit for primary prevention in lower-risk people, the real-world rate of muscle symptoms outside of controlled trials, and how much all-cause mortality improves specifically in low-risk primary-prevention groups as opposed to higher-risk populations overall.

Everything else this guide covers, insulin sensitivity, inflammation, liver health, blood pressure drivers beyond salt, sleep, nutrition, and movement, is complementary work either way. It addresses the underlying terrain that produced the numbers your prescriber is treating, alongside that conversation, not instead of it.

Where this leaves your prescription. None of the above is a reason to distrust your prescriber or the medication itself, and it is also not a reason to treat "everyone should be on a statin" as beyond discussion. If you have documented cardiovascular disease, a prior heart attack or stroke, or a stent or bypass, the evidence for staying on your statin exactly as prescribed is strong and not seriously contested. This guide has nothing to add to that conversation except to keep taking it. If you have not had one of those events and are looking at a rising number for the first time, the honest, evidence-based position is that this is a real, individual decision with a genuinely small absolute benefit at the lower end of risk, one worth a specific conversation with your prescriber rather than an automatic yes or an automatic no. Either way, any decision about starting, stopping, changing, or adjusting a statin, a blood pressure medication, or a blood thinner belongs to you and your prescribing physician, based on your actual risk category and follow-up numbers, never to a supplement company, a podcast, or this guide.

If your functional-medicine work moves your numbers meaningfully over time (which is genuinely possible and is the entire point of addressing the terrain), that same rule applies: any change to dosing or discontinuation is a conversation to have with your prescribing physician, never a decision to make unilaterally or based on how you feel. Bring your ApoB, Lp(a), and triglyceride-to-HDL numbers from Key 2 to that conversation, along with where you fall on the primary-versus-secondary-prevention picture above. They give your prescriber a sharper, more complete basis for that specific decision, not a reason to second-guess what has already been prescribed.

The same logic applies to blood pressure medications and blood thinners. ACE inhibitors, ARBs, beta blockers, calcium channel blockers, diuretics, and anticoagulants each treat a specific, documented mechanism, and stopping or reducing any of them without your prescriber's involvement carries real risk, including the stroke and cardiac emergencies described earlier in this guide.

KEY 4

Inflammation and the Health of Your Blood Vessel Lining

Where plaque actually starts. Plaque does not build up in a healthy, smooth artery. It builds up where the single layer of cells lining the inside of your blood vessels, the endothelium, has already been disrupted. That disruption is called endothelial dysfunction, and it typically develops years before any cholesterol number or blood pressure reading looks abnormal.

How the vessel lining breaks down. A healthy endothelium produces nitric oxide, a signaling molecule that keeps blood vessels relaxed, discourages clotting, and calms inflammation locally. Sustained exposure to the risk factors this guide covers (insulin resistance, elevated blood pressure, oxidative stress, smoking) reduces nitric oxide availability, and endothelial dysfunction is now recognized as an early, reversible precursor of atherosclerosis, the pathophysiological link between early risk factors and the plaque that eventually causes heart attacks and strokes.[12] That word, reversible, describes this specific, early cellular process, not a promise about an existing diagnosis.

High-sensitivity C-reactive protein, or hs-CRP, is a blood marker of that inflammatory process, produced by the liver in response to inflammatory signaling elsewhere in the body. It adds prognostic information on cardiovascular risk that is comparable in strength to blood pressure or cholesterol themselves, and current risk categories treat a level under 1 mg/L as lower risk, 1 to 3 mg/L as average, and above 3 mg/L as higher relative risk.[11] In a 30-year study following more than 27,000 initially healthy women, hs-CRP, LDL cholesterol, and Lp(a) each independently predicted long-term cardiovascular events, with hs-CRP carrying the single strongest association of the three.[13] Inflammation is not a side note to the cholesterol story. In this data, it was the most powerful of the three markers measured.

What to ask your physician for. hs-CRP is a standard, inexpensive blood test your physician can order, and it is worth asking for if it has not already been checked. Read it for what it is: a nonspecific screening marker. It tells you an inflammatory load exists somewhere. It does not tell you where that load is coming from, and it confirms nothing on its own. A recent infection, an injury, a flaring joint, or a dental problem will all raise it, which is why it is interpreted alongside your other numbers and repeated when you are not acutely ill. What lowers it tends to be the same terrain-level work covered throughout this guide: addressing insulin resistance, visceral fat, sleep, and the nutrition and movement patterns in Key 9, all of which independently calm this inflammatory signal.

KEY 5

Your Liver Is a Metabolic Organ, Not Just an Alcohol Story

An outdated assumption. For decades, a fatty liver on imaging was assumed to be an alcohol problem, and people who did not drink heavily were often confused or dismissive when the finding showed up on their own scan. That assumption is outdated. In most cases in the United States today, a fatty liver has nothing to do with alcohol at all.

The name change, and why it happened. The condition long called non-alcoholic fatty liver disease (NAFLD) was formally renamed in 2023 by a multi-society consensus of liver specialists from 56 countries. The new name is metabolic dysfunction-associated steatotic liver disease, or MASLD, and the new diagnostic definition requires the presence of at least one of five cardiometabolic risk factors (excess weight, elevated blood pressure, elevated triglycerides, low HDL, or elevated blood sugar), replacing the old approach of diagnosing it only by ruling out alcohol.[14] The name change reflects what the field already understood mechanistically: this is fundamentally the liver's expression of the same insulin resistance driving Key 1, not primarily a toxin exposure story. Chronically elevated insulin drives the liver to convert excess circulating fuel into fat and store it directly inside liver cells, and over time that stored fat drives local inflammation and, in a meaningful share of cases, progressive scarring (fibrosis).

This matters well beyond the liver itself. A meta-analysis of 36 studies following more than 5.8 million people found that having a fatty liver was associated with a 45 percent higher rate of fatal or non-fatal cardiovascular events compared with not having one, a risk that climbed further, to roughly two and a half times higher, in people with more advanced fibrosis, independent of age, sex, adiposity, diabetes, and other standard cardiometabolic risk factors.[15] Practically, that means a fatty liver finding on an ultrasound is not an isolated footnote. It is a signal worth treating as seriously as an abnormal lipid panel, because in this data, it carried independent cardiovascular weight of its own.

What actually helps the liver. The liver responds well to the same terrain-level changes that improve insulin sensitivity elsewhere: reduced intake of refined carbohydrate and added sugar, regular movement, and, where relevant, weight change through the lifestyle patterns covered in Key 9. Elevated liver enzymes (ALT, AST, and GGT) on a standard metabolic panel are inexpensive early clues worth tracking over time rather than dismissing as a minor lab flag, and they belong on the worksheet at the end of this guide alongside your lipid and blood pressure numbers.

KEY 6

Blood Pressure Is Bigger Than Salt

More than one lever. “Cut your salt” is the most common piece of advice given for high blood pressure, and for some people it genuinely helps. But it is far from the whole story, and treating sodium as the only lever leaves several other well-documented drivers unaddressed.

What raises pressure besides sodium. As covered in Key 1, chronically elevated insulin itself promotes sodium retention in the kidney and increases sympathetic nervous system activity, both of which raise blood pressure independent of how much salt is actually on your plate. Two minerals matter more than most people realize: a dose-response meta-analysis of potassium supplementation trials found a real blood pressure-lowering effect, strongest in people who already have hypertension and in those with high sodium intake, though the relationship is U-shaped rather than “more is always better,” with the benefit leveling off and reversing at very high intakes, which is exactly why potassium status is worth assessing with your physician rather than self-supplementing at high doses, particularly if you take an ACE inhibitor, ARB, or potassium-sparing diuretic, or have any kidney impairment.[16] Magnesium supplementation shows a similar pattern in a 2025 meta-analysis of 38 randomized trials: a modest average reduction in blood pressure across all participants, but a substantially larger reduction, roughly seven to eight points systolic, specifically in people already being treated for hypertension or who had low magnesium status to begin with.[17] Read together, both minerals appear to matter most for the people who need the most help, not as a blanket fix for everyone.

Two other drivers deserve more attention than they usually get. Obstructive sleep apnea, repeated pauses in breathing during sleep that fragment rest and spike stress hormones overnight, is strongly linked to resistant hypertension, the kind that does not respond well to standard medication combinations. A meta-analysis found people with obstructive sleep apnea were roughly three to four times more likely to have resistant hypertension than people without it.[18] If your blood pressure is difficult to control despite adherence to medication, snoring, witnessed pauses in breathing, or unrefreshing sleep are worth mentioning to your physician as a possible undiagnosed contributor. Chronic psychological stress also shows a real, if still-debated, association with hypertension risk, with one meta-analysis finding people under chronic psychosocial stress were roughly two and a half times more likely to have hypertension, while the authors themselves noted the underlying studies vary enough in how they define and measure stress that more research is needed to firm up the relationship.[19] Kidney function matters too, in both directions: the kidneys and the blood pressure system regulate each other through the renin-angiotensin-aldosterone pathway, so impaired kidney filtration can itself drive blood pressure up, which is part of why a basic metabolic panel checking kidney markers belongs alongside a blood pressure log.

The DASH evidence, and what to ask for. Beyond sodium reduction, the DASH dietary pattern (rich in fruits, vegetables, and low-fat dairy, with reduced saturated fat) has one of the strongest evidence bases of any dietary blood pressure intervention: the original randomized trial found it lowered systolic and diastolic blood pressure by 5.5 and 3.0 points respectively compared with a typical diet, with a substantially larger 11.4 and 5.5 point reduction in the subgroup who already had hypertension.[20] Ask your physician about checking your magnesium and potassium status, mention sleep quality and snoring at your next visit, and consider a home blood pressure log (the worksheet at the end of this guide includes one) so patterns show up before your next appointment rather than only in a single office reading.

KEY 7

Visceral Fat, Why Where You Store It Matters

Same weight, different risk. Two people can carry the same total body weight and the same BMI and have meaningfully different cardiometabolic risk, because it is not just how much fat you carry, it is where you carry it.

Why location changes the risk. Fat stored around and inside the abdominal organs, visceral fat, behaves entirely differently from fat stored just under the skin. A comprehensive review of the physiology describes visceral fat as part of a broader pattern, including dysfunctional fat cell expansion and fat stored in places it does not belong (the liver and muscle), that is closely tied to the clustering of cardiometabolic risk factors covered throughout this guide: elevated triglycerides, increased free fatty acid release into the bloodstream, release of inflammatory signaling molecules directly from fat tissue, liver insulin resistance, small dense LDL particles, and reduced HDL.[21] In practical terms, visceral fat is not passive storage. It is an active, inflammatory tissue that manufactures many of the same downstream problems described in Keys 1 through 6.

This is also why waist circumference and waist-to-hip ratio carry real, independently documented predictive value. The INTERHEART study, a standardized case-control study spanning 52 countries and more than 27,000 participants, found abdominal obesity (measured by waist-to-hip ratio) was one of just nine modifiable factors that together accounted for roughly 90 to 94 percent of the population-level risk of a first heart attack worldwide, alongside smoking, an elevated ApoB-to-ApoA1 ratio (the same particle-count logic from Key 2, which on its own accounted for the second-largest share of risk after smoking), high blood pressure, diabetes, psychosocial stress, and diet and activity patterns.[22] That is a global, cross-cultural dataset confirming that where you store fat, not just how much you weigh, is one of the small handful of factors that explain most heart attack risk worldwide.

What to track, and why it moves first. A simple waist circumference measurement, taken at the navel, adds real information beyond the number on the scale or your BMI, and it is worth tracking over time on the worksheet at the end of this guide. The encouraging part of this research is that visceral fat responds preferentially to the same lifestyle interventions in Key 9. When people lose weight through nutrition and movement changes, visceral fat tends to mobilize first, ahead of fat stored elsewhere.

KEY 8

Sleep and Circadian Rhythm, the Underrated Driver

Treated as a nicety, not a driver. Sleep is treated as a lifestyle nicety in most cardiovascular conversations, something to improve “if you have time.” The evidence does not support that ranking. Sleep duration and timing function as a direct cardiometabolic input, not an afterthought.

What the data actually shows. A systematic review and meta-analysis following nearly 475,000 people found that short sleep duration was associated with a 48 percent higher risk of developing or dying from coronary heart disease and a 15 percent higher stroke risk, and, somewhat counterintuitively, long sleep duration (generally more than eight or nine hours) was also associated with increased risk across coronary heart disease, stroke, and total cardiovascular disease, suggesting the relationship is U-shaped rather than “more sleep is always better.”[23] Shift work adds a separate, related problem: circadian misalignment, the mismatch between your internal biological clock and your actual sleep and eating schedule, is itself an independent risk factor for overweight, obesity, type 2 diabetes, elevated blood pressure, and metabolic syndrome, distinct from sleep duration alone.[24] The mechanism runs back through the same insulin pathway covered in Key 1: circadian disruption impairs glucose regulation and insulin sensitivity even when total sleep time looks adequate on paper.

What to prioritize. Consistent sleep and wake times, even on weekends, appear to matter as much as total sleep duration. If your schedule includes rotating or overnight shifts, that is a genuine, independent cardiometabolic risk factor worth naming explicitly to your physician, not just a scheduling inconvenience. The worksheet at the end of this guide includes a simple place to log typical sleep hours alongside your other numbers, since patterns over weeks tell you more than any single night.

KEY 9

Nutrition and Movement, What Actually Has Evidence Behind It

Noisy advice, specific answers. Nutrition advice for cardiometabolic risk is noisy, contradictory, and often reduced to "eat less, move more" without much specificity. Several interventions in this space have real, randomized evidence behind them, and it is worth knowing which ones.

What the trials actually tested. A Mediterranean-pattern diet, emphasizing olive oil, vegetables, legumes, fish, and nuts, has been tested directly against a standard low-fat diet advice approach in a large Spanish randomized trial of over 7,400 adults at high cardiovascular risk. After correcting for early protocol irregularities at some study sites and reanalyzing the full dataset, the corrected results still showed roughly a 30 percent reduction in major cardiovascular events (heart attack, stroke, and cardiovascular death combined) in the groups assigned to a Mediterranean diet supplemented with either extra-virgin olive oil or mixed nuts, compared with the reduced-fat control diet.[25] Since this guide reports funding everywhere it matters: that trial was funded by Spain's public health research institute, with the olive oil and nuts supplied as donations from olive oil and nut industry groups which, per the published report, had no role in the design or analysis of the study.[25] On the movement side, a randomized trial directly measuring insulin sensitivity found that a single session of either moderate-intensity continuous exercise or high-intensity interval training produced roughly a 20 percent improvement in insulin sensitivity the very next day compared with the pre-training baseline, with both exercise styles working equally well, though the benefit faded within about four days without another session, which is exactly why consistency matters more than any single "best" workout format.[26] On the fiber side, a meta-analysis of 28 randomized trials found that at least 3 grams daily of oat beta-glucan, a soluble fiber, lowered LDL cholesterol by an average of about 0.25 mmol/L (roughly 10 mg/dL) without changing HDL or triglycerides.[27] None of these numbers are dramatic in isolation. Together, sustained over time, they represent a genuinely evidence-based lever most people never hear quantified.

The three concrete levers. None of this requires a rigid or extreme diet. The practical takeaway is specific rather than general: a Mediterranean-style eating pattern most days, movement most days of the week (even moderate-intensity movement shows the same insulin-sensitizing effect as harder interval training), and a deliberate daily source of soluble fiber (oats, legumes, and psyllium are the best-studied sources) are three concrete, evidence-supported levers, not vague lifestyle suggestions. These changes work on the same insulin resistance root cause from Key 1, which is why they tend to move cholesterol, blood pressure, and liver fat together rather than requiring three separate diets for three separate numbers.

Your Staged Path Forward

Everything above describes the problem and the general categories of solution. Actually working through it in your own case benefits from a structured, staged approach rather than trying to change everything at once.

Stage 1: Get the full picture. Beyond your standard lipid panel and a single blood pressure reading, ask your physician about ApoB, a one-time Lp(a), hs-CRP, fasting insulin, homocysteine, and liver enzymes (ALT, AST, GGT), and start logging blood pressure at home using the worksheet below. You cannot address what you have not measured.

Stage 2: Address the foundational terrain. For a defined trial period, commonly eight to twelve weeks, focus on the nutrition pattern, movement, sleep consistency, and stress covered in Keys 6 through 9, while continuing every prescribed medication exactly as directed. This stage is about the modifiable terrain, not about judging or replacing what your prescriber already started.

Stage 3: Reassess and personalize. Repeat the labs and blood pressure log from Stage 1, and bring the results to your prescribing physician. This is the point where genuinely individualized decisions get made, including whether targeted nutrient support (magnesium, potassium, fiber) makes sense for your specific numbers, whether a sleep study is warranted, and whether any medication adjustment is appropriate, a decision that belongs to your prescriber based on your actual data.

Stage 4: Maintain and recheck periodically. Cardiometabolic risk is a long arc, not a single fix. Periodic rechecking, roughly every three to six months initially, keeps the picture current as your terrain continues to change.

Your Numbers Worksheet

Tracking your own numbers over time is one of the most genuinely useful things you can do with this information. Bring this log to every appointment.

Blood pressure and vitals

Date	Systolic	Diastolic	Resting heart rate	Weight	Waist circumference

Labs (repeat roughly every 3 to 6 months, or as your physician directs)

Draw date	ApoB	LDL-C	Lp(a) (once is usually enough)	Triglycerides	HDL	TG/HDL ratio	hs-CRP	Homocysteine	Fasting glucose	HbA1c	Fasting insulin	ALT / AST / GGT

Fasting insulin sits in that row next to glucose and HbA1c on purpose. Insulin climbs for years to hold glucose in range, so a blood sugar workup that measures only the two downstream numbers finds the problem late. Ask for all three together.

Sleep and lifestyle notes

Week of	Average sleep hours	Shift work this week?	Movement days	Notable stress

Take the Next Step: Your Free Root Discovery Session

The Root Discovery Session is a free consultation with me to review your specific numbers: what your labs and blood pressure actually show, which drivers from this guide look most relevant in your case, and what a personalized, prescriber-coordinated plan could look like. There is no obligation, and if you decide to move forward, this conversation is simply the first step.

Consultations and ongoing care are both available wherever you are located. Being outside North Carolina is not a barrier to working with me. Tell me where you are when you reach out and I will lay out exactly how we would work together from there.

Schedule your free Root Discovery Session at rootcauseunlocked.com/discovery.

You can also reach the office directly at 919-662-0520.

If you would rather start on your own, start tracking.

The numbers worksheet earlier in this guide works on paper. The digital version does the same job on your phone, keeps your history in one place, and lets you watch the numbers that actually move: home blood pressure readings over weeks rather than one anxious reading in an exam room, and how they respond as you change things. Bringing that history to your next lab review makes it dramatically more useful than a single point in time.

Start your free tracker at rootcauseunlocked.com/tracker.

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References

Every citation below was independently retrieved and confirmed against its live PubMed record (esearch plus esummary and, for every citation, full abstract confirmation via efetch) before use. None are drawn on trust from any prior document, including the practice's own internal protocol guides. Full verification detail, including several claims investigated and left uncited or unused, is in the companion file Heart-Health-Citation-Ledger.md.

1. Nolan CJ, Prentki M. Insulin resistance and insulin hypersecretion in the metabolic syndrome and type 2 diabetes: Time for a conceptual framework shift. Diabetes & Vascular Disease Research. 2019 Mar;16(2):118-127. PMID 30770030. https://pubmed.ncbi.nlm.nih.gov/30770030/
2. Chen W, Wang S, Lv W, Pan Y. Causal associations of insulin resistance with coronary artery disease and ischemic stroke: a Mendelian randomization analysis. BMJ Open Diabetes Research & Care. 2020 May;8(1):e001217. PMID 32398352. https://pubmed.ncbi.nlm.nih.gov/32398352/
3. Huang X, Zhao JV. Exploring the pathways linking fasting insulin to coronary artery disease: a proteome-wide Mendelian randomization study. BMC Medicine. 2025 May 30;23(1):321. PMID 40442727. https://pubmed.ncbi.nlm.nih.gov/40442727/
4. Zhao JV, Luo S, Schooling CM. Sex-specific Mendelian randomization study of genetically predicted insulin and cardiovascular events in the UK Biobank. Communications Biology. 2019;2:332. PMID 31508506. https://pubmed.ncbi.nlm.nih.gov/31508506/
5. Richardson TG, Sanderson E, Palmer TM, Ala-Korpela M, Ference BA, Davey Smith G, Holmes MV. Evaluating the relationship between circulating lipoprotein lipids and apolipoproteins with risk of coronary heart disease: A multivariable Mendelian randomisation analysis. PLoS Medicine. 2020 Mar;17(3):e1003062. PMID 32203549. https://pubmed.ncbi.nlm.nih.gov/32203549/
6. Deng Y, Zhang C, Wang J, Zeng J, Zhang J, Zhang T, Zhao H, Li M, Zhao Y, Gan W, Shao Y, Yu H, Zhou W, Zhang C. Application of serum pools in insulin harmonization: Commutability and stability. Annals of Clinical Biochemistry. 2023 May;60(3):199-207. PMID 36750430. https://pubmed.ncbi.nlm.nih.gov/36750430/
7. Glavinovic T, Thanassoulis G, de Graaf J, Couture P, Hegele RA, Sniderman AD. Physiological Bases for the Superiority of Apolipoprotein B Over Low-Density Lipoprotein Cholesterol and Non-High-Density Lipoprotein Cholesterol as a Marker of Cardiovascular Risk. Journal of the American Heart Association. 2022 Oct 18;11(20):e025858. PMID 36216435. https://pubmed.ncbi.nlm.nih.gov/36216435/
8. Reyes-Soffer G, Ginsberg HN, Berglund L, Duell PB, Heffron SP, Kamstrup PR, Lloyd-Jones DM, Marcovina SM, Yeang C, Koschinsky ML; American Heart Association Council on Arteriosclerosis, Thrombosis and Vascular Biology. Lipoprotein(a): A Genetically Determined, Causal, and Prevalent Risk Factor for Atherosclerotic Cardiovascular Disease: A Scientific Statement From the American Heart Association. Arteriosclerosis, Thrombosis, and Vascular Biology. 2022 Jan;42(1):e48-e60. PMID 34647487. https://pubmed.ncbi.nlm.nih.gov/34647487/
9. Baneu P, Vacarescu C, Dragan SR, Cirin L, Lazar-Hocher AI, Cozgarea A, Faur-Grigori AA, Crisan S, Gaita D, Luca CT, Cozma D. The Triglyceride/HDL Ratio as a Surrogate Biomarker for Insulin Resistance. Biomedicines. 2024 Jul 5;12(7):1493. PMID 39062066. https://pubmed.ncbi.nlm.nih.gov/39062066/
10. Cholesterol Treatment Trialists' (CTT) Collaboration; Fulcher J, O'Connell R, Voysey M, Emberson J, Blackwell L, Mihaylova B, Simes J, Collins R, Kirby A, Colhoun H, Braunwald E, La Rosa J, Pedersen TR, Tonkin A, Davis B, Sleight P, Franzosi MG, Baigent C, Keech A. Efficacy and safety of LDL-lowering therapy among men and women: meta-analysis of individual data from 174,000 participants in 27 randomised trials. Lancet. 2015 Apr 11;385(9976):1397-1405. PMID 25579834. https://pubmed.ncbi.nlm.nih.gov/25579834/
11. Ridker PM. A Test in Context: High-Sensitivity C-Reactive Protein. Journal of the American College of Cardiology. 2016 Feb 16;67(6):712-723. PMID 26868696. https://pubmed.ncbi.nlm.nih.gov/26868696/
12. Mudau M, Genis A, Lochner A, Strijdom H. Endothelial dysfunction: the early predictor of atherosclerosis. Cardiovascular Journal of Africa. 2012 May;23(4):222-231. PMID 22614668. https://pubmed.ncbi.nlm.nih.gov/22614668/
13. Ridker PM, Moorthy MV, Cook NR, Rifai N, Lee IM, Buring JE. Inflammation, Cholesterol, Lipoprotein(a), and 30-Year Cardiovascular Outcomes in Women. New England Journal of Medicine. 2024 Dec 5;391(22):2087-2097. PMID 39216091. https://pubmed.ncbi.nlm.nih.gov/39216091/
14. Rinella ME, Lazarus JV, Ratziu V, Francque SM, Sanyal AJ, Kanwal F, Romero D, Abdelmalek MF, Anstee QM, Arab JP, et al; NAFLD Nomenclature consensus group. A multisociety Delphi consensus statement on new fatty liver disease nomenclature. Journal of Hepatology. 2023 Dec;79(6):1542-1556. PMID 37364790. https://pubmed.ncbi.nlm.nih.gov/37364790/
15. Mantovani A, Csermely A, Petracca G, Beatrice G, Corey KE, Simon TG, Byrne CD, Targher G. Non-alcoholic fatty liver disease and risk of fatal and non-fatal cardiovascular events: an updated systematic review and meta-analysis. Lancet Gastroenterology & Hepatology. 2021 Nov;6(11):903-913. PMID 34555346. https://pubmed.ncbi.nlm.nih.gov/34555346/
16. Filippini T, Naska A, Kasdagli MI, Torres D, Lopes C, Carvalho C, Moreira P, Malavolti M, Orsini N, Whelton PK, Vinceti M. Potassium Intake and Blood Pressure: A Dose-Response Meta-Analysis of Randomized Controlled Trials. Journal of the American Heart Association. 2020 Jun 16;9(12):e015719. PMID 32500831. https://pubmed.ncbi.nlm.nih.gov/32500831/
17. Argeros Z, Xu X, Bhandari B, Harris K, Touyz RM, Schutte AE. Magnesium Supplementation and Blood Pressure: A Systematic Review and Meta-Analysis of Randomized Controlled Trials. Hypertension. 2025 Nov;82(11):1844-1856. PMID 41000008. https://pubmed.ncbi.nlm.nih.gov/41000008/

18. Ahmed AM, Nur SM, Xiaochen Y. Association between obstructive sleep apnea and resistant hypertension: systematic review and meta-analysis. Frontiers in Medicine. 2023 Jun 2;10:1200952. PMID 37332747. https://pubmed.ncbi.nlm.nih.gov/37332747/

19. Liu MY, Li N, Li WA, Khan H. Association between psychosocial stress and hypertension: a systematic review and meta-analysis. Neurological Research. 2017 Jun;39(6):573-580. PMID 28415916. https://pubmed.ncbi.nlm.nih.gov/28415916/

20. Appel LJ, Moore TJ, Obarzanek E, Vollmer WM, Svetkey LP, Sacks FM, Bray GA, Vogt TM, Cutler JA, Windhauser MM, Lin PH, Karanja N. A clinical trial of the effects of dietary patterns on blood pressure. DASH Collaborative Research Group. New England Journal of Medicine. 1997 Apr 17;336(16):1117-1124. PMID 9099655. https://pubmed.ncbi.nlm.nih.gov/9099655/

21. Tchernof A, Despres JP. Pathophysiology of human visceral obesity: an update. Physiological Reviews. 2013 Jan;93(1):359-404. PMID 23303913. https://pubmed.ncbi.nlm.nih.gov/23303913/

22. Yusuf S, Hawken S, Ounpuu S, Dans T, Avezum A, Lanas F, McQueen M, Budaj A, Pais P, Varigos J, Lisheng L; INTERHEART Study Investigators. Effect of potentially modifiable risk factors associated with myocardial infarction in 52 countries (the INTERHEART study): case-control study. Lancet. 2004 Sep 11-17;364(9438):937-952. PMID 15364185. https://pubmed.ncbi.nlm.nih.gov/15364185/

23. Cappuccio FP, Cooper D, D'Elia L, Strazzullo P, Miller MA. Sleep duration predicts cardiovascular outcomes: a systematic review and meta-analysis of prospective studies. European Heart Journal. 2011 Jun;32(12):1484-1492. PMID 21300732. https://pubmed.ncbi.nlm.nih.gov/21300732/

24. Hemmer A, Mareschal J, Dibner C, Pralong JA, Dorribo V, Perrig S, Genton L, Pichard C, Collet TH. The Effects of Shift Work on Cardio-Metabolic Diseases and Eating Patterns. Nutrients. 2021 Nov 22;13(11):4178. PMID 34836433. https://pubmed.ncbi.nlm.nih.gov/34836433/

25. Estruch R, Ros E, Salas-Salvado J, Covas MI, Corella D, Aros F, Gomez-Gracia E, Ruiz-Gutierrez V, Fiol M, Lapetra J, et al; PREDIMED Study Investigators. Primary Prevention of Cardiovascular Disease with a Mediterranean Diet Supplemented with Extra-Virgin Olive Oil or Nuts. New England Journal of Medicine. 2018 Jun 21;378(25):e34 (corrected and republished from 2013;368(14):1279-1290). PMID 29897866. https://pubmed.ncbi.nlm.nih.gov/29897866/

26. Ryan BJ, Schleh MW, Ahn C, Ludzki AC, Gillen JB, Varshney P, Van Pelt DW, Pitchford LM, Chenevert TL, Gioscia-Ryan RA, Howton SM, Rode T, Hummel SL, Burant CF, Little JP, Horowitz JF. Moderate-Intensity Exercise and High-Intensity Interval Training Affect Insulin Sensitivity Similarly in Obese Adults. Journal of Clinical Endocrinology and Metabolism. 2020 Aug 1;105(8):e2941-e2959. PMID 32492705. https://pubmed.ncbi.nlm.nih.gov/32492705/

27. Whitehead A, Beck EJ, Tosh S, Wolever TM. Cholesterol-lowering effects of oat beta-glucan: a meta-analysis of randomized controlled trials. American Journal of Clinical Nutrition. 2014 Dec;100(6):1413-1421. PMID 25411276. https://pubmed.ncbi.nlm.nih.gov/25411276/

28. Ference BA, Ginsberg HN, Graham I, Ray KK, Packard CJ, Bruckert E, Hegele RA, Krauss RM, Raal FJ, Schunkert H, et al. Low-density lipoproteins cause atherosclerotic cardiovascular disease. 1. Evidence from genetic, epidemiologic, and clinical studies. A consensus statement from the European Atherosclerosis Society Consensus Panel. European Heart Journal. 2017 Aug 21;38(32):2459-2472. PMID 28444290. https://pubmed.ncbi.nlm.nih.gov/28444290/

29. Scandinavian Simvastatin Survival Study Group. Randomised trial of cholesterol lowering in 4444 patients with coronary heart disease: the Scandinavian Simvastatin Survival Study (4S). Lancet. 1994 Nov 19;344(8934):1383-1389. PMID 7968073. https://pubmed.ncbi.nlm.nih.gov/7968073/

30. Cholesterol Treatment Trialists' (CTT) Collaborators; Mihaylova B, Emberson J, Blackwell L, Keech A, Simes J, Barnes EH, Voysey M, Gray A, Collins R, Baigent C. The effects of lowering LDL cholesterol with statin therapy in people at low risk of vascular disease: meta-analysis of individual data from 27 randomised trials. Lancet. 2012 Aug 11;380(9841):581-590. PMID 22607822. https://pubmed.ncbi.nlm.nih.gov/22607822/

31. Ridker PM, Danielson E, Fonseca FA, Genest J, Gotto AM Jr, Kastelein JJ, Koenig W, Libby P, Lorenzatti AJ, MacFadyen JG, et al; JUPITER Study Group. Rosuvastatin to prevent vascular events in men and women with elevated C-reactive protein. New England Journal of Medicine. 2008 Nov 20;359(21):2195-2207. PMID 18997196. https://pubmed.ncbi.nlm.nih.gov/18997196/

32. de Lorgeril M, Salen P, Abramson J, Dodin S, Hamazaki T, Kostucki W, Okuyama H, Pavy B, Rabaeus M. Cholesterol lowering, cardiovascular diseases, and the rosuvastatin-JUPITER controversy: a critical reappraisal. Archives of Internal Medicine. 2010 Jun 28;170(12):1032-1036. PMID 20585068. https://pubmed.ncbi.nlm.nih.gov/20585068/

33. Lu Y, Zhou S, Dreyer RP, Spatz ES, Geda M, Lorenze NP, D'Onofrio G, Lichtman JH, Spertus JA, Ridker PM, Krumholz HM. Sex Differences in Inflammatory Markers and Health Status Among Young Adults With Acute Myocardial Infarction: Results From the VIRGO Study. Circulation: Cardiovascular Quality and Outcomes. 2017 Feb;10(2):e003470. PMID 28228461. https://pubmed.ncbi.nlm.nih.gov/28228461/

34. Sattar N, Preiss D, Murray HM, Welsh P, Buckley BM, de Craen AJ, Seshasai SR, McMurray JJ, Freeman DJ, Jukema JW, et al. Statins and risk of incident diabetes: a collaborative meta-analysis of randomised statin trials. Lancet. 2010 Feb 27;375(9716):735-742. PMID 20167359. https://pubmed.ncbi.nlm.nih.gov/20167359/

35. Cholesterol Treatment Trialists' (CTT) Collaboration; Reith C, Preiss D, Blackwell L, Emberson J, Spata E, Davies K, Halls H, Harper C, Holland L, Wilson K, et al. Effects of statin therapy on diagnoses of new-onset diabetes and worsening glycaemia in large-scale randomised blinded statin trials: an individual participant data meta-analysis. Lancet Diabetes & Endocrinology. 2024 May;12(5):306-319. PMID 38554713. https://pubmed.ncbi.nlm.nih.gov/38554713/

36. Newman CB, Preiss D, Tobert JA, Jacobson TA, Page RL 2nd, Goldstein LB, Chin C, Tannock LR, Miller M, Raghuveer G, et al; American Heart Association Clinical Lipidology, Lipoprotein, Metabolism and Thrombosis Committee. Statin Safety and Associated Adverse Events: A Scientific Statement From the American Heart Association. Arteriosclerosis, Thrombosis, and Vascular Biology. 2019 Feb;39(2):e38-e81. PMID 30580575. https://pubmed.ncbi.nlm.nih.gov/30580575/

37. Herrett E, Williamson E, Brack K, Beaumont D, Perkins A, Thayne A, Shakur-Still H, Roberts I, Prowse D, Goldacre B, et al; StatinWISE Trial Group. Statin treatment and muscle symptoms: series of randomised, placebo controlled n-of-1 trials. BMJ. 2021 Feb 24;372:n135. PMID 33627334. https://pubmed.ncbi.nlm.nih.gov/33627334/

38. Homocysteine Studies Collaboration. Homocysteine and risk of ischemic heart disease and stroke: a meta-analysis. JAMA. 2002 Oct 23-30;288(16):2015-2022. PMID 12387654. https://pubmed.ncbi.nlm.nih.gov/12387654/

39. Clarke R, Bennett DA, Parish S, Verhoef P, Dotsch-Klerk M, Lathrop M, Xu P, Nordestgaard BG, Holm H, Hopewell JC, et al; MTHFR Studies Collaborative Group. Homocysteine and coronary heart disease: meta-analysis of MTHFR case-control studies, avoiding publication bias. PLoS Medicine. 2012 Feb;9(2):e1001177. PMID 22363213. https://pubmed.ncbi.nlm.nih.gov/22363213/

40. Li Y, Huang T, Zheng Y, Muka T, Troup J, Hu FB. Folic Acid Supplementation and the Risk of Cardiovascular Diseases: A Meta-Analysis of Randomized Controlled Trials. Journal of the American Heart Association. 2016 Aug 15;5(8):e003768. PMID 27528407. https://pubmed.ncbi.nlm.nih.gov/27528407/

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